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DETAIL SPECIFICATION

TILE, REFRACTORY, SILICON CARBIDE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers silicon carbide refractory tile for use as baffles and tube alignment in naval boiler furnaces.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Other Government documents, drawings, and publications. The following other Government documents, drawing, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-1385936 - Tile, Baffle and Alignment, Silicon Carbide for Marine Boilers

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVSEA PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via TDMIS at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST Online database at https://assist.dla.mil.

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2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS 411-1 - Hazardous Material Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspections by Attributes

(Copies of this document are available online at www.webstore.ansi.org)

ASTM INTERNATIONAL

ASTM C20 - Standard Test Methods for Apparent Porosity, Water Absorption, Apparent Specific Gravity, and Bulk Density of Burned Refractory Brick and Shapes by Boiling Water

ASTM C863 - Standard Test Method for Evaluating Oxidation Resistance of Silicon Carbide Refractories at Elevated Temperatures

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 21068-1 - Chemical analysis of silicon-carbide-containing raw materials and refractory products – Part 1: General information and sample preparation

ISO 21068-2 - Chemical analysis of silicon-carbide-containing raw materials and refractory products – Part 2: Determination of loss on ignition, total carbon, free carbon and silicon carbide, total and free silica and total and free silicon

ISO 21068-3 - Chemical analysis of silicon-carbide-containing raw materials and refractory products – Part 3: Determination of nitrogen, oxygen and metallic and oxidic constituents

(Copies of these documents are available online at www.iso.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2 and 6.3.

3.2 Material. The tile shall be composed of an oxidic (clay) bonded silicon carbide material formed to a uniform, dense structure.

3.3 Bulk density. The bulk density shall be not less than 156.07 pounds per cubic foot (2.50 grams per cubic centimeter) (see 4.5.1).

3.4 Apparent porosity. The apparent porosity shall be not greater than 18 percent (see 4.5.1).

3.5 Chemical composition. The silicon carbide content shall be not less than 85 percent by weight of the finished product. The product shall contain not greater than 0.5 percent of lime (CaO), not greater than 0.25 percent of alkalies, and not greater than 0.1 percent of magnesia (MgO) (see 4.5.2.2).

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3.6 Oxidation resistance. Tile shall have a protective glaze that is formed by the mix of constituents as part of the normal production process for the product offered. Where heavy surface glazes are used to improve resistance, they shall be fired after application and create no health hazard for handlers. Tiles shall be supplied dry.

3.7 Dimensions. Tile shall be furnished in the size and shape specified (see 6.2) and shall conform to the dimensions and tolerances shown on drawing 803-1385936 (see 4.5.3).

3.8 Marking. Each tile shall be marked with the piece number and the manufacturer's brand name or symbol in accordance with drawing 803-1385936 (see 4.5.3).

3.9 Workmanship. The silicon carbide tile shall be of uniform structure, bulk density, and apparent porosity throughout with no cracks, voids, or soft centers. The tile shall have smooth dense surfaces. Edges and corners shall be sufficiently solid and strong to prevent crumbling and chipping during normal handling (see 4.5.3).

3.10 Toxicity and prohibited materials.

3.10.1 Toxicity. When evaluated in accordance with 4.7, the tiles shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.7 and 6.4).

3.10.2 Prohibited materials. The tile shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)

b. Conformance inspection (see 4.3)

4.2 First article inspection. First article inspection shall be performed on the tiles when a first article sample is required (see 3.1). The inspection shall include the examinations and tests shown in table I. Examination of the end item shall be made in accordance with the inspection levels and classification of defects shown in tables II and III.

TABLE I. First article and conformance inspections.

Characteristic	First article	Conformance	Requirement	Verification
Material		X	3.2	Table III
Bulk density	X	X	3.3	4.5.1
Apparent porosity	X	X	3.4	4.5.1
Chemical composition	X	X	3.5	4.5.2.2
Oxidation resistance	X		3.6	4.6
Dimensions	X	X	3.7	4.5.3
Marking		X	3.8	4.5.3
Workmanship		X	3.9	4.5.3
Toxicity		X	3.10.1	4.7
Prohibited materials		X	3.10.2	4.8

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TABLE II. Sampling for first article and conformance inspection.

Inspection	Test	Inspection level
Material and workmanship	Table III	S-4
Marking	Table III	S-1
Bulk density	4.5.1	S-3
Apparent porosity	4.5.1	S-3
Chemical composition	4.5.2.2	1/
Dimensions	Table III	S-4
Count and weight of tiles	Table III	S-1

NOTES:

1/ Chemical composition inspection should consist of five tiles selected at random. An average of two determinations for each constituent shall be reported.

TABLE III. Classification of defects.

Categories	Defects
Critical	None defined.
Major	
101	Cracked, broken laminations. Surfaces, not smooth or dense. Grainy, coarse appearance due to non-refractory grade silicon carbide grain or improper grinding and mixing constituents.
102	Corner or edges chipped or crumbled affecting serviceability.
103	Any specified dimension not within the tolerances specified on Drawing 803-1385936.
104	Omitted, illegible, incorrect, incomplete.
Minor	
201	Number of tiles per container less than indicated requirements.
202	Gross weight exceeds specified requirements.

4.3 Conformance inspection. Conformance inspection shall consist of the examinations and tests shown in table I.

4.4 Sampling. Sampling for first article and conformance inspection shall be in accordance with ANSI/ASQ Z1.4 at the inspection level specified in table II and the Acceptance Quality Limit (AQL) specified (see 6.2). For purposes of sampling, an inspection lot for examinations and tests shall consist of all material of the same type offered for delivery at one time. No less than five sample units shall be randomly selected throughout the lot. Where two-piece assemblies are produced, samples of both pieces shall be included. No more than one sample shall be drawn from one container. Unless otherwise specified (see 6.2), the unit of sample shall be one tile.

4.5 Test procedures.

4.5.1 Bulk density and apparent porosity. The bulk density and apparent porosity shall be determined in accordance with ASTM C20. Samples shall be tested intact without cutting or breaking. Results shall be recorded numerically to nearest 0.1 pound per cubic foot for bulk density and 0.1 percent for apparent porosity.

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4.5.2 Chemical analysis.

4.5.2.1 Sample pretreatment. Sampling and preparation of the test samples shall be done in accordance with ISO 21068-1. Samples prepared in this manner from each sample unit shall be combined, thoroughly mixed and quartered. From the composite sample, duplicate samples shall be taken for two parallel chemical analyses.

4.5.2.2 Chemical analysis procedure. Chemical analysis, to establish compliance with 3.5, shall be in accordance with ISO 21068-2 for determination of silicon carbide content. The determination of carbon is to be based on oxygen extraction. The indirect qualitative method shall be used. Total carbon shall be determined by induction furnace combustion with the BAM S003 as the Certified Reference Material (CRM). Detection of the released carbon dioxide (CO_2) shall be in accordance with ISO-21068-2 infrared absorption. Free carbon shall be determined in a resistance furnace with quartz tube combustion chamber with indirect detection methods. Analysis of oxides (CaO, MgO, and alkalies shall be done in accordance with any of the four methods described in ISO 21068-3). The average of the two determinations for each constituent shall be reported. Calculate the test results as a composition (mass fraction) expressed as a percentage. If an integer part of the mass fraction is two digits, express the result to one decimal place. When the integer part of the mass fraction is one digit record the result to two decimal places.

4.5.3. Visual examination. The tile shall be visually and dimensionally examined to verify conformance to this specification. Deviation from the specified material, dimensions, marking, and workmanship shall be cause for rejection.

4.6 Oxidation resistance. Maximum volume change when tested in accordance with ASTM C863 at 1,832 °F (1,000 °C) for 500 hours shall be 1.5 percent (see 6.11).

4.7 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.10.1 as specified (see 6.2). The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the tile using data provided by the manufacturer/distributor to the NMCPHC (see 3.10.1 and 6.4).

4.8 Prohibited materials. Prohibited materials shall be verified as specified (see 3.10.2 and 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The silicon carbide refractory tile is intended for use as baffle tile (deflector), tube spacer tile, and tube alignment tile in naval boiler furnaces. The tile is used in naval boilers built in accordance with MIL-B-18381. This specification has cyclic operation and shock-hardened brickwork requirements that are not found in commercial boilers. The types and grades of refractory products suitable for this service has been determined by testing. Current stock of tile under National Stock Numbers (NSNs), as shown on drawing 803-1385936 Piece Numbers 1 through 10 and 953, may be used until depleted; future requisitions for tile Piece Numbers 12 through 16 and 953 should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article inspection is required (see 3.1 and 6.3).

c. Size and shape of tile required (see 3.7).

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d. Requirements for an HHA (see 3.10.1 and 4.7).

e. Requirement for prohibited materials verification (see 3.10.2 and 4.8).

f. AQL of 2.5 and when an AQL of 2.5 is acceptable for workmanship and dimensional defects (see 4.4).

g. Packaging requirements (see 5.1 and 6.8).

h. Additional requirements for Certificates of Compliance (see 6.5 and 6.6). Reference to Navy Data Item Description (DI-MISC-81356 or similar data requirement for content and format).

i. When Safety Data Sheets (SDS) submission and distribution is required (see 6.7).

j. Level of preservation and packing requirement (see 6.8.1 and 6.9).

k. When shipping containers or pallets may contain other than tile of one size and shape only (see 6.8.1 and 6.9).

l. Special marking required (see 6.10).

6.3 First article. When a first article inspection is required (see 6.2), the item should be a first article sample. The first article should consist of six tiles. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that Government approval is presently appropriate for the pending contract. A waiver can be recommended based on the contract award being less than 3 years from date of previous acceptable first article testing for a particular brand name, and certification from the actual manufacturer that there has been no change in tile manufacturing location, fabrication process, composition, or source of basic heat resistant raw materials since the earlier test, and no discontinuation of production for an extended period of time. The contracting officer can permit waiver beyond the 3-year interval when there has been no discontinuation of production, and minor changes of factors mentioned above are certified by the manufacturer to not affect performance of the tile.

6.4 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, https://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health hazard-assessment.aspx.

6.5 Certificates of compliance. The certificates of compliance should include the results of verification testing for each requirement, as applicable. This will include the found test results, the specified requirements, and the conclusion that each result passed. Additional requirements of the certificates of compliance content or format may be specified (see 6.2). A copy of each certificate should be provided to Commanding Officer, NSWC Philadelphia, Attn: Code 412, 5001 South Broad St., Philadelphia PA 19112, in addition to other reporting requirements of the contract.

6.6 Attached certification. Consideration should be given to including certificates of compliance (see 6.5) with each shipment of tile. Certificates should indicate successful completion of the individual tests of the verification inspections as described in Section 4, or state when previous testing was performed that was the basis for any waived testing requirements.

6.7 SDSs. When required (see 6.2), contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. The pertinent government mailing addresses for submission of data are listed in FED-STD-313. In order to obtain the SDS, FAR clause 52.223-3 must be in the contract.

6.8 Government packaging. When specified (see 6.2), the packaging requirements specified herein, 6.8.1 through 6.10 should apply only for direct Government acquisitions in place of the packaging requirements in 5.1.

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6.8.1 Preservation. Preservation should be level A, military pack level minimum, or commercial, as specified (see 6.2). Unless otherwise specified (see 6.2), packs should contain tile of one size (PC. No.) and type.

a. Level A. Tile should be packed in fiberboard boxes in accordance with ASTM D5118/D5118M, weather-resistant (WR). Boxes should be of variety DW (double wall) or variety SW (single wall). SW boxes should be provided with a full height inside liner of the same grade as the box used. Tiles should be individually wrapped with cushioning media or segregated through the use of fiberboard partitions. Additional cushioning, as required, should be used to fill all void spaces within the boxes. Loose fill material including straw or sawdust should not be used. Boxes should be closed and sealed in accordance with closure method 5D3 and sealing method B of ASTM D1974/D1974M. Gross weight of the boxes should not exceed 40 pounds.

b. Military pack level minimum. Tile should be packed as specified in 6.8.1.a except that boxes should be of the domestic (D) class, minimum grade 200, and box closures should be in accordance with closure method 1A of ASTM D1974/D1974M.

c. Commercial. Tile should be packed in accordance with ASTM D3951.

6.9 Packing. Packing should be level A, B, military pack level minimum, or commercial as specified (see 6.2). Unless otherwise specified (see 6.2), shipping containers or pallets should contain packs of tile of one size (PC. No.) and type only.

a. Level A. Tile, preserved as specified in 6.8.1, should all be packed in wood or wood cleated plywood boxes in accordance with ASTM D6880/D6880M Class 1 Style 1 or 2 (plywood), ASTM D6251 overseas type or unitized on pallets in accordance with MIL-STD-147, load type 1a, bonding means “O” or “P” respectively. Box closure and reinforcement should be in accordance with the box specification. Gross weight of wood boxes should not exceed 200 pounds of palletized loads 3,000 pounds.

b. Level B. Tile, preserved as specified in 6.8.1, should be packed as specified in 6.8.1.a, except that wood and wood cleated plywood boxes may be of the domestic type of class (without water-repellant treatment).

c. Military pack level minimum. Tile, preserved as specified in 6.8.1, may be shipped without over packing. When specified (see 6.2), preserved tiles should be in unit loads on pallets in accordance with MIL-STD-147, with load type and bonding means at the option of the contractor.

d. Commercial. Tile, preserved as specified in 6.8.1, should be packed in accordance with ASTM D3951.

6.10 Marking. In addition to any special marking required (see 6.2), level A, B and military pack level minimum interior and exterior boxes and palletized unit loads should all be marked in accordance with MIL-STD-129 and include bar code markings. Commercial interior and exterior boxes and palletized unit loads should all be marked in accordance with ASTM D3951 and should all include bar code marking of MIL-STD-129.

6.11 Submission of oxidation resistance test data. Since the ASTM C863 test is not intended as a referee test, and there is a large critical difference in clay bonded silicon carbide testing averages, consideration will be given to products that exceed the 1.5-percent volume increase requirement specified in 3.6. In such cases, complete test data that includes discussion of micro crack formation, weight change, and decrease in open porosity should be submitted.

6.12 Subject term (key word) listing.

Apparent porosity

Bulk density

Deflector

Heat resistant

Oxide (oxidic)

Tube alignment

Tube baffle

6.13 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – SH

Preparing activity:
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